

CSMMagTech 2026 Rare-Earth-Free Magnet Research

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Materials Research

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Niron Fe16N2 performance	24 MGOe demonstrated	27 MGOe on production samples (2025)	12.5% BHmax improvement changes substitution feasibility
EU CRMA compliance	Not in V1.0	CRMA 2024: 40% EU/allied RE by 2030; recycling mandate	Legal compliance gap in European market operations
MXene-NdFeB magnet	Not in V1.0	Drexel 2025: +12% coercivity, +25degC Tc with MXene additive	New option for high-temp NdFeB partially replacing SmCo
Fe-Sn intermetallic	Not known	NIMS Japan 2025: 12 MGOe demonstrated	New RE-free candidate for low-cost applications
Magnet costs	Dy spot \$150/kg	Dy \$4,100/kg May 2026 (China ECA restriction)	V1.0 magnet cost estimates 27x too low for Dy-containing grades

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMMagTech 2026 Rare-Earth-Free Magnet Research | June 2026